



Does human–AI collaboration lead to more creative art? Aesthetic evaluation of human-made and AI-generated haiku poetry

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ABSTRACT

With the development of technology, the quality of AI-generated text has improved. This is relevant in the AI art field, where AI generates literature or poetry that is appreciated. This study compared human-made and AI-generated haiku poetry, which is composed with 17 syllables and the world's shortest and clearest rules, to examine aesthetic evaluations of AI art and people's beliefs about it. AI-generated haiku were divided into those with and without human intervention. Two tasks were completed by 385 participants. The first involved evaluating human-made and AI-generated haiku on 21 items, such as beauty. The second involved determining whether the haiku were human-made or AI-generated. The results showed that the beauty rating of the AI-generated haiku with the human intervention was the highest, and those of the human-made and AI-generated haiku without human intervention were equal. Furthermore, participants could not distinguish between human-made and AI-generated haiku. These results suggest that human–AI collaboration has better creativity in haiku production. Moreover, a negative correlation was found between discrimination performance and beauty rating in AI-generated haiku, suggesting that high-quality AI-generated work is believed to be human-made. This study indicates the potential of human–AI collaboration in haiku and the underestimation of AI art due to algorithm aversion.

1. Introduction

Artificial Intelligence (AI) art is flourishing. The main focus of AI art has been on visual arts such as painting (Elgammal et al., 2017; Gatys et al., 2016; Karayev et al., 2013; Li & Wand, 2016, see Cetinic & She, 2022 for review). Recently, the dramatic improvement in natural language processing and generation technology has enabled AI to create literature and poetry that closely resemble those created by humans. This kind of AI art has attracted the attention of researchers in the fields of computer science and robotics on the generation side, and those from psychology and philosophy, where the debate centers on understanding beauty and creativity (c.f., Daniele & Song, 2019).

Previous studies evaluating AI art have compared algorithmic and human-made art, asking participants to discriminate the artist (human vs. AI) and to evaluate aesthetic scores (e.g., Chamberlain et al., 2018; Gangadharbatla, 2022; Ueda et al., 2021). Artist discrimination applies the Turing test to AI art, which verifies that the capabilities of a machine or artificial intelligence are equal to, or indistinguishable from, the

intellectual activities performed by humans. Studies on painting show consistent results that the discrimination between human-made and AI-generated paintings is difficult (Chamberlain et al., 2018; Elgammal et al., 2017; Gangadharbatla, 2022; Ragot, Martin, & Cojean, 2020; Ueda et al., 2021). Moreover, Chamberlain et al. (2018) found that the discrimination was also difficult when the participants were art majors. Interestingly, even if they cannot distinguish the artists of paintings, when asked to rate their preference and beauty, the scores differed (Ragot et al., 2020; Ueda et al., 2021, but it was not found in Chamberlain et al., 2018). Moreover, some studies showed that the preference in art is not influenced by the actual authorship but rather whom a person believes the author to be (i.e., attributional authorship, Chamberlain et al., 2018; Ragot et al., 2020, but not in Hong & Curran, 2019).

Given that AI can now generate literature and poetry, in recent years, it is worth extending the discussion to the linguistic arts. Studies comparing human-made and AI-generated literature and poetry has been examined since the late 2010s (Gunser et al., 2022; Hopkins & Kiela, 2017; Köbis & Mossink, 2021; Lau et al., 2018; Lc, 2021; Oliveira,

Abbreviations: HOTL, Human-out-of-the-loop; HITL, Human-in-the-loop; LSTM, Long short-term memory; IRI, Interpersonal Reactivity Index.

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2009; Schwartz, 2015; see Table 1 for the summary). Oscar Schwartz first demonstrated whether it was possible to distinguish between the two (Schwartz, 2015). When his “Bot or Not” website asked viewers to compare human-made and algorithmically generated poems, 65% of algorithmically generated poems deceived the author (i.e., poems written by AI seemed to have been written by humans). Later, Lau et al. (2018) generated 14-line sonnets by Deep-speare consisting of language, rhythm, and rhyme models, and asked laypersons and an expert of English literature to evaluate them. Although the laypeople could not distinguish between AI-generated and human-made work, the expert judged the former inferior to the latter regarding readability and evoked emotions. Note that in the study by Lau et al. (2018), there was only one expert, and thus, it was unclear whether other experts would also be able to distinguish between them or whether this differentiation was unique to this particular expert.

In these studies, AI-generated poems were ultimately selected by humans; however, in order to make human involvement in AI poems more rigorous, Köbis and Mossink (2021) distinguished AI poems into human-out-of-the-loop (HOTL) and human-in-the-loop (HITL), where the former did not require human intervention when selecting items from AI-generated poems, while the latter required stimuli to be generated by AI and selected by humans. The results showed that AI poems with HOTL could be distinguished from human-made ones, but those with HITL were not. In other words, the findings revealed that human involvement could increase the quality of AI poetry, making it equivalent to that of humans. More recently, Gunser et al. (2022) replicated Köbis and Mossink (2021) using more materials and found that participants could not distinguish between human and AI poems with HITL. At the same time, human poems were found to be superior in multiple evaluations, such as *inspiring* and *well-written*.

The present study attempts to extend the previous findings to the world’s shortest poem, the haiku. Haiku, which originated in Japan, has clear rules, such as a fixed form of 5-7-5 syllables and the inclusion of seasonal words called “*kigo*” (Iida, 2008). These rules of haiku make it different from other forms of poetry. The first aspect is ambiguity; since the number of characters in haiku is small, the expression is ambiguous, and readers have to supplement and interpret the scene from the little information they have (Hitsuwari & Nomura, 2022a; 2022c). Therefore, even though AI generates ambiguous haiku, readers will attempt to interpret them according to the scenes they can imagine. The second is that the impression of haiku depends on one or two mental images evoked (Blasko & Merski, 1998; Hitsuwari & Nomura, 2022b). Other types of poetry are generally longer than haiku and often translate to various images depending on different scenes, which eliminates the problem of having consistency and a sense of narrative between scenes. Despite the difficulty of composing haiku with a small number of characters, AI has attempted this task (Kawamura et al., 2021; Yokoyama et al., 2019).¹ These characteristics make AI-generated haiku highly evaluated and difficult to distinguish from human-generated ones.

In this study, we further examined the factors that contributed to the evaluation of beauty in human-made and AI-generated haiku. Previous studies evaluated poems with some items (e.g., readability and emotional evocation in Hopkins & Kiela, 2017, and well-written, inspiring, fascinating, interesting, and aesthetic in Gunser et al., 2022; details in Table 1). Although recent studies on neuroaesthetics suggest that aesthetic is composed of sensory-motor, emotional-valuation, and meaning-knowledge systems (Chatterjee & Vartanian, 2014), the number of items used in previous studies was too small to examine the

factors that evoke the feeling of beauty in haiku. To determine the general characteristics of the beauty experience across object kinds, Brielmann et al. (2021) proposed eleven dimensions that have been considered by prominent philosophers of aesthetics (pleasure, wishing to continue the experience, feeling alive, feeling that the experience is beautiful to everyone, number of felt connections to the experience, longing, feeling free of desire, mind wandering, surprise, wanting to understand the experience more, and feeling that the experience tells a story) and eight dimensions conveyed by psychologists (complexity, arousal or excitement, learning from the experience, wanting to understand, harmony in variety, meaningfulness, exceeding one’s expectation, and interest).² In accordance with Brielmann et al. (2021), these dimensions were used to identify factors that delineate the experience of beauty in human-made and AI-generated haiku. In addition, we exploratory examined feelings of awe and nostalgia, which have been involved in a previous haiku study as well (Hitsuwari & Nomura, 2022c).

A problematic psychological characteristic in the evaluation of AI art is algorithm aversion, the belief that beautiful things are made by humans, not AI (Burton et al., 2020). This is also associated with a preference for AI poetry (Köbis & Mossink, 2021). Given that Okanda et al. (2019) showed that animism and empathy traits influenced the evaluation of moral aspects of robots, individual differences in these traits may influence the impressions of robots and AI, and could be related to the discrimination between human-made and AI-generated haiku.

2. Hypotheses

This study aimed to examine how haiku, as a literary art form created by AI, humans, and a collaboration between them, affects beauty scores and discrimination of authors. Our main concerns were fourfold: while the first two focused on the evaluation of the work, the latter two focused more on the attributional authorship.

First, we compared the beauty scores of human-made, AI-generated without human intervention (i.e., HOTL), and AI-generated with human intervention (i.e., HITL) haiku. Since human involvement likely increases the quality of AI poetry (e.g., Köbis & Mossink, 2021), HITL haiku would be more evaluated than HOTL. Moreover, AI-generated haiku may be evaluated the same as, or even better than, human-made haiku, since people can sufficiently engage their imagination, even with little information.

Second, we examined the factors that can explain the beauty of human-made and AI-generated (HOTL/HITL) haiku. Although factors affecting the aesthetic perception of human-made and AI-generated art are diverse, no study of AI poetry has addressed using a copious number of evaluation items. Following the previous study (Brielmann et al., 2021), we expected that aesthetic evaluations are associated with both philosophical and psychological perspectives.

Third, we examine whether participants can distinguish between human-made and AI-generated (HOTL/HITL) haiku. Referring to the previous study using typical poems (Köbis & Mossink, 2021), we expect that participants would be able to distinguish between HOTL and human haiku but not between HITL and human haiku. However, given that haiku have less information and more reliance on the reader’s imagination than other poems, even HOTL haiku may not be distinguished from human haiku.

Finally, we examine which factors, especially participants’ backgrounds (education and experience of haiku) and personality traits (i.e., animism and empathy, which are associated with attitudes toward robots and AI), explain the discrimination accuracy of authors. Based on the finding of Chamberlain et al. (2018), which stated that author

¹ In a TV show demonstration (not an empirical study) that evaluated the quality of five human-made and AI-generated haiku each, the highest rated haiku was the AI-generated one although the results of the one-on-one comparisons showed that human-made haiku had a higher win rate than AI-generated ones.

² For a detailed explanation of these dimensions, please see the original text and references in Brielmann et al. (2021).

Table 1

Literature review comparing human-made and AI-generated poetry.

ID	Study	HITL or HOTL	AI algorithm	Human poems	Number of evaluators	Number of poems	Rating	Discriminating	Limitations	Notes
1	Schwartz (2015)	HITL	Various algorithms (e.g., RACTER & RKCP)	Professional poets (e.g., William Blake & Frank O'Hara)	Thousands of people	NA	NA	- 65% of them could not identify the author	- Not empirical study (without control of experimental conditions or participants)	- First attempt to examine whether human and AI poetry can be discriminated
2	Hopkins and Kiela (2017)	HITL	LSTM	Classic poets (e.g., Shakespeare)	70	Eight manmade and two AI-generated poems	- Human poems were rated slightly higher quality (in total score of form, readability, and emotional evocation) than AI-generated poems - The poems judged to be the most human and aesthetic were AI-generated	- 48.6% of human poems were falsely attributed to AI and 53.8% of AI poems were falsely attributed to humans	- Small sample size of poems and participants	
3	Crowdworkers in Lau et al. (2018)	HOTL	Deep-seare	Professional poets (e.g., Shakespeare)	1000	50 manmade and 180 AI-generated quatrains	NA	- Accuracy was 53.2%, indicating it was hard to distinguish between human and AI poems	- No statistical tests were performed	- Each participant rated a few poems
4	Expert in Lau et al. (2018)	HOTL	Deep-seare	Professional poets (e.g., Shakespeare)	1	30 manmade and 90 AI-generated quatrains	- AI-generated poems was better in meter and rhyme than human-made. - Human-made poems were better in emotion and readability than AI-generated	NA	- Only one evaluator	- Evaluator was an expert in English literature
5	Lc (2021)	HITL	GPT-2	Author made	25	5 manmade and 5 AI-generated poems	- Evaluation regarding structure was not significantly different, but expression was rated higher for human-made than AI-generated poems - Human-made poems were rated higher than AI-generated poems with a 57.0% probability.	- Accuracy was 61.6% for human-made poems and 56.0% for AI-generated poems, but there was no significant difference between them	- Small sample size of poems and participants	- the AI could reproduce the nuance of the original text
6	Study 1 in Köbis and Mossink (2021)	HITL	GPT-2	Novice made	192 (About half of them participated in the discrimination task)	20 manmade and ten AI-generated poems	- Human-made poems were rated higher than AI-generated poems with a 57.0% probability.	- 50.2% of all answers correctly discriminated the author	- Only HITL	
7	Study 2 in Köbis and Mossink (2021)	HITL & HOTL	GPT-2	Classic poets (Maya Angelou & Hermann Hesse)	384 (185 participated in the discrimination task)	Ten manmade, ten AI-generated with HITL, and ten AI-generated with HOTL poems	- Participants chose the better of the human-made and AI-generated poems, and human-made poetry was	- 65.5% in the HOTL poems and 53.7% in the HITL poems could be discriminated	- Little variety in human-made poems	- First experiment comparing HOTL and HITL

(continued on next page)

Table 1 (continued)

ID	Study	HITL or HOTL	AI algorithm	Human poems	Number of evaluators	Number of poems	Rating	Discriminating	Limitations	Notes
							higher evaluated in 64.9% than AI-generated. (62.9% for HITL and 66.9% for HOTL)			
8	Study 1 in Gunser et al. (2022)	HITL	GPT-2	Experts (Gunser et al., 2022)	120	18 manmade and 18 AI-generated with HITL poems	- Human-made poems were rated higher in 5 evaluations (well-written, inspiring, fascinating, interesting, and aesthetic) than AI-generated	- 40.3% of human-made poems were falsely attributed to AI, and 42.0% of AI-generated poems were falsely attributed to human-made	- Only HITL	- Expert poets and GPT-2 created continuations of the classical poems
9	Study 2 in Gunser et al. (2022)	HITL	GPT-2	Classic poets (Franz Kafka, Friedrich Hölderlin, Robert Gernhardt, & Paul Celan)	302	18 manmade and 18 AI-generated with HITL poems	- Human-made poems were rated higher in five evaluations (well-written, inspiring, fascinating, interesting, and aesthetic) than AI-generated	- 33.5% of human-made poems were falsely attributed to AI, and 40.2% of AI-generated poems were falsely attributed to human-made	- Human-made poems could be easily detected due to their historical language usage and sublime style	

discrimination accuracy did not change depending on whether the participant was an art major or not, it was expected that the participants' backgrounds would not significantly affect the accuracy of discrimination between human-made and AI-generated ones. Moreover, it was expected that animism and empathy, which decrease algorithmic aversion, would work to the advantage of author discrimination, since algorithmic aversion instantly attributes a work of beauty to a human authorship.

3. Methods

This study was approved by the institutional review board of the Psychological Science Unit at Kyoto University (2-P-21). All data, materials, and scripts are available online (<https://osf.io/s6wny/>).

3.1. Participants

Assuming a small effect size, 322 subjects would be needed to compare the three ways of the generated haiku (i.e., human-made, HITL vs. HOTL). Therefore, we conducted a web-based experiment to recruit 400 Japanese participants through CrowdWorks (<https://crowdworks.jp>). We excluded 15 participants who did not respond appropriately to an attention check which involved asking for a specific answer in the questionnaire. Therefore, data from 385 participants (191 male and 194 female participants, $M_{age} = 40.9$, $SD = 10.1$) were included in the final analysis. All participants were informed of the study's purpose, methodology, risks, duration of the experiment, the voluntary nature of participation, their right to withdraw, and how the participants information was to be handled. Furthermore, written informed consent was obtained before they participated in the experiment. They were paid 500 JPY for their participation.

3.2. Materials

3.2.1. Haiku stimulus

To disperse the seasons and genres of haiku, we selected ten seasonal words that must be included in haiku. Forty human-made haiku (four for each of the ten seasonal words) were selected from the Saijiki, which contains several haiku poems created by professionals for each seasonal word. All of haiku in the Saijiki are highly regarded by the professionals. In a preliminary survey, eight laypersons (not included in the actual pool of participants) confirmed that they had never seen these haikus before. For AI-generated haiku, we used the haiku generation system based on the Long Short-Term Memory (LSTM) algorithm developed by the Harmonic System Engineering Lab at Hokkaido University, Japan (Kawamura et al., 2021; Yokoyama et al., 2019). The system first generates a set of candidate haiku sentences using a language model trained on haiku data by LSTM. The system then applies morphological analysis to the generated sentences and selects those that match the form of seasonal fixed form haiku. The similarity between the generated sentences and the haiku in the training data is calculated and sentences that are more similar than a certain threshold are removed. Depending on the frequency of seasonal words, the number of generated haiku varies. For each of the ten seasonal words, the system generated 36,442 to 624,130 haikus. In the algorithm, the AI itself calculated the degree of their validity as Japanese language and finally, generated the top 500 haiku as AI-generated haiku.

From the AI-generated haiku, we randomly selected 20 (two for each of the ten seasonal words) and designated them as HOTL haiku. Another 20 haiku (two for each of the ten seasonal words) that were highly rated for their beauty by three amateur/novice haiku artists were designated HITL haiku.

The 80 haiku were then randomly divided into two stimulus lists of 40 haiku, each consisting of 20 human-made, ten HOTL, and ten HITL haiku per list.

3.2.2. Questionnaires to investigate personality traits

To examine the effect of personality traits on judgments, we measured personality traits using five types of questionnaires.

First, the Interpersonal Reactivity Index (IRI) (Davis, 1980; Himichi et al., 2017) was used to measure trait empathy. The scale consists of four subscales: empathic concern, personal distress, perspective-taking, and fantasy, with a five-point Likert scale.

Second and third, the Adult Animism Scale (Ikeuchi, 2010) and the Aliveness Animism Scale (Okanda et al., 2019) were used to measure animism traits. The former is an 11-item scale with a five-point Likert scale consisting of three subscales: the deification of natural objects, the alterity of the owner, and anthropomorphism of possessions. This scale reflects the tendency to perceive divinity and the existence of life in inanimate objects, even though they do not possess life. The latter scale reflects the tendency to think that inanimate objects are alive and asks the participants to select all the objects that they think are alive from eight items (lit candle, telephone, clock, doll, teddy bear, banana peel, cloud, and mud) and four plants (tree, flower, grass, and vegetable).

Fourth, the following questions were used to measure knowledge of and interest in haiku and art: haiku experience ("I just learned about it in elementary school or junior high school," "I have had few opportunities to experience it as an adult," "I have had opportunities to experience it as an adult," "I have had many opportunities to experience it as an adult," "I have had opportunities to experience it every day as an adult."), frequency of museum visits ("more than twice a month," "about once a month," "a few times a year," "about once a year," "I rarely go," "I have never been"), experience in creative jobs ("no such experience," "a little," "a lot," "quite a lot"), and interest in the arts ("not at all," "very little," "a little," "a lot," "quite a lot").

Fifth, in addition to the above, as exploratory research, a four-item scale developed by Sugimori and Kusumi (2014) was used to measure the frequency of déjà vu experiences and sensitivity to similarity. Two items measured the frequency of the déjà vu experience using a seven-point Likert scale (1: did not have the experience in the past year to 7: every day), and the other two items measured the degree to which the déjà vu experience applied to them using a five-point Likert scale (1: definitely does not apply to 5: definitely applies to me).

3.3. Procedure

The participants accessed the experiment page available on the Internet. They first read the informed consent information, and started the experiment upon agreeing to it. The experiment consisted of three blocks: the *rating block* to evaluate the haiku, the *discriminating block* to judge whether the creator of the haiku was human or not, and the *trait block* to measure the personality traits of the participants. Following Chamberlain et al. (2018), the influence of author attribution (prior knowledge about whether the work produced by AI was included in the list) was controlled with half the participants completing the *rating block* first, followed by the *discriminating block*. The other half completed the *discriminating block* first, followed by the *rating block*. For both sets of participants, the *trait block* was completed at the end of the experiment. Furthermore, both sets of participants were presented with two different haiku lists (including 40 haikus, each), to investigate the consistency of the results in both lists.

Rating block. Here, human-made and AI-generated haiku were presented individually. Participants were asked to evaluate them using the following 21 dimensions on a 7-point Likert scale: *psychological factors* related to aesthetic evaluations such as beauty, valence, arousal, awe, empathy, vividness, passion, novelty, nostalgia, and déjà vu, and certain *philosophical factors* including pleasure, wish to continue the experience, alive, universality, number of connections, longing, free of desire, mind wandering, surprise, want to understand, and tells story (see Briemann et al., 2021 for the details).

Discriminating block. Here, the haiku were presented and the participants were asked to judge whether each was created by humans

or AI. After judging all of the haiku, they chose from the following 12 items that provided clues for their decisions: the rhythm of words, consistency, regularity, repeatability, complexity, depth, abstraction, intentionality, uniqueness, expression, nuance, and others. Finally, they were asked to provide an open-ended explanation of their decision of choosing the haikus as human or AI-generated.

Trait block. Here, in addition to age, sex, educational background, and nationality, the participants were asked to answer the five types of trait questionnaires. Educational backgrounds were categorized as: (1) junior high school, (2) high school, (3) junior college/technical college, (4) university, and (5) graduate school.

3.4. Data analysis

For the first purpose, we averaged each person's beauty score for each group and compared them with one-factor ANOVA (human haiku, HOTL haiku, vs. HITL haiku) using the "anovakun" function (ver. 4.8.6; Iseki, 2021) in R (ver. 4.1.0). Furthermore, we conducted the same analysis on the 20 ratings other than beauty for exploratory analysis.

Second, we examined whether the 20 ratings other than beauty could explain the beauty score with a linear mixed model using the lmer package (Bates et al., 2015). A total of 15,400 observations, with 385 individuals and 40 haikus, met the sample size of the literature (Arend & Schäfer, 2019) on the testing power of hierarchical data and repeated measures data (called multilevel models). The dependent variable was the beauty score, the independent variables were 20 ratings other than beauty and haiku conditions, and educational background was the control variable. The participants, haiku, and task order (whether the rating block or the discriminating block was first) were put into random effects. The independent variables were centered within a cluster. The educational background was included as a control variable because prior knowledge can significantly influence the evaluation of haiku (Sato, 2007).

Third, to examine whether participants could distinguish between human and AI haiku, we conducted a *t*-test to see if the hit rate significantly differed from the chance level (0.5).

Finally, to examine whether individual traits could explain the discrimination accuracy, we analyzed a generalized linear mixed model (logistic analysis) with hit or not as the dependent variable, individual traits as the independent variable, and participants and haiku as random effects. The independent variables were centered using the grand mean.

4. Results

4.1. Beauty scores for human-made and AI-generated haiku

Table 2 shows the descriptive statistics for each item of the haiku evaluations. We compared the beauty scores of the human, HOTL, and HITL conditions (i.e., the first row of Table 2). The result showed that the main effect of condition was $F(384, 2) = 212.41, p = .00, \eta^2 = 0.06$. Following multiple comparisons showed that HITL had the highest scores. There was no difference in the scores between the human and HITL conditions (see Fig. 1). Table 2 also shows the results of exploratory analyses on other 20 ratings. These results suggest that AI-generated haiku were more highly evaluated than the human-made ones when humans intervened in the AI outputs (i.e., HITL).

4.2. Explanatory factors for beauty of human-made and AI-generated haiku

Next, we determined the most explanatory factors for the aesthetic evaluation by examining whether the 20 factors concerning the beauty evaluation, task order, haiku list, and condition could explain the beauty of haiku using a linear mixed model (Table 3). All factors related to the beauty evaluation had contributions to the beauty of haiku, except alive ($b = -0.01, SE = 0.01, t = -0.56, p = .57$), mind wandering ($b = 0.01,$

Table 2

Descriptive statistics for rating items of Haiku evaluation.

	Mean (SD)			F value	p value	η^2	Multiple comparison
	Human	HOTL	HITL				
Beauty	4.15 (.75)	4.14 (.78)	4.56 (.74)	212.41	.00	.06	Human = HOTL < HITL
Déjà vu	3.30 (1.06)	3.17 (1.05)	3.50 (1.09)	106.63	.00	.02	HOTL < human < HITL
Image	4.60 (.68)	3.99 (.79)	4.63 (.75)	302.23	.00	.13	HOTL < human = HITL
Valence	4.11 (.58)	4.00 (.60)	4.28 (.64)	92.19	.00	.04	HOTL < human < HITL
Arousal	3.96 (.69)	3.82 (.74)	4.08 (.74)	75.88	.00	.02	HOTL < human < HITL
Awe	3.39 (.99)	3.36 (1.02)	3.68 (1.03)	110.72	.00	.02	Human = HOTL < HITL
Nostalgia	4.14 (.89)	4.08 (.96)	4.28 (.90)	42.19	.00	.01	HOTL < human < HITL
Novelty	3.63 (.70)	3.57 (.73)	3.64 (.77)	6.70	.00	.00	HOTL < human = HITL
Empathy	3.81 (.87)	3.53 (.92)	3.97 (.89)	181.03	.00	.04	HOTL < human < HITL
Intention	4.48 (.76)	4.35 (.81)	4.59 (.78)	54.67	.00	.02	HOTL < human < HITL
Joy	3.40 (.83)	3.31 (.87)	3.57 (.87)	68.85	.00	.02	HOTL < human < HITL
Continue	3.80 (1.00)	3.71 (1.05)	4.01 (1.00)	92.46	.00	.02	HOTL < human < HITL
Alive	3.58 (.99)	3.44 (1.02)	3.73 (1.02)	78.92	.00	.01	HOTL < human < HITL
Universality	3.82 (.77)	3.79 (.80)	4.17 (.77)	177.46	.00	.05	Human = HOTL < HITL
Longing	3.58 (.97)	3.49 (1.04)	3.71 (.99)	46.68	.00	.01	HOTL < human < HITL
Free desire	3.34 (.92)	3.43 (.95)	3.53 (.98)	29.93	.00	.01	Human < HOTL < HITL
Mind wandering	3.20 (1.00)	3.20 (1.02)	3.39 (1.02)	50.12	.00	.01	Human = HOTL < HITL
Connection	3.34 (1.02)	3.31 (1.05)	3.47 (1.03)	31.06	.00	.00	Human = HOTL < HITL
Surprise	3.06 (1.04)	2.95 (1.04)	3.07 (1.06)	22.74	.00	.00	HOTL < human = HITL
Understand	4.10 (.97)	4.15 (1.03)	4.27 (1.00)	28.67	.00	.01	HOTL < human < HITL
Tells story	4.44 (.83)	4.41 (.86)	4.60 (.84)	44.50	.00	.01	Human = HOTL < HITL

**Fig. 1.** Beauty scores in the human, HOTL, and HITL conditions.

$SE = 0.01$, $t = 1.17$, $p = .24$), and connection ($b = -0.01$, $SE = 0.01$, $t = -1.31$, $p = .19$). Task order ($b = -0.04$, $SE = 0.08$, $t = -0.42$, $p = .67$), haiku list ($b = -0.06$, $SE = 0.06$, $t = -1.04$, $p = .30$), and the conditional difference between human and HOTL ($b = 0.11$, $SE = 0.09$, $t = 1.35$, $p = .18$) did not explain the beauty of haiku, while the conditional difference between human and HITL ($b = 0.18$, $SE = 0.09$, $t = 2.16$, $p = .03$) did. As shown in Section 4.1, these results showed that AI-generated haiku with

human intervention (i.e., HITL) had a higher beauty score than human-made haiku, and the score of AI-generated haiku without human intervention (i.e., HOTL) was compatible with human-made haiku. The result also indicated that task order (i.e., prior knowledge about whether the work produced by AI) did not affect the evaluation of the beauty of haiku. Furthermore, the result did not depend on the stimulus set that was presented to the participants. Refer to [Supplementary Table 1](#) for the results of the linear mixed model for each condition.

4.3. Distinguishing between human-made and AI-generated haiku

For the attributional authorship (i.e., the participants' abilities to distinguish between human-made and AI-generated haiku), hit rates (i.e., the probability of correctly judging human-made haiku as human-made and AI-generated haiku as AI-generated) for the human, HOTL, and HITL conditions were .55, .50, and .43, respectively ([Fig. 2](#); see also [Supplementary Table 2](#) for the hit rate of each haiku and list). Although the hit rate in the human condition was significantly higher than the chance level, $t(39) = 3.51$, $p = .001$, in the HITL condition it was significantly lower than the chance level, $t(19) = -3.19$, $p = .005$. The hit rate in the HOTL condition was not different from the chance, $t(19) = 0.03$, $p = .98$. These results suggest that the participants could not distinguish between human-made and AI-generated for HOTL haiku. Furthermore, they believed that the authorship of HITL haiku was human rather than AI.

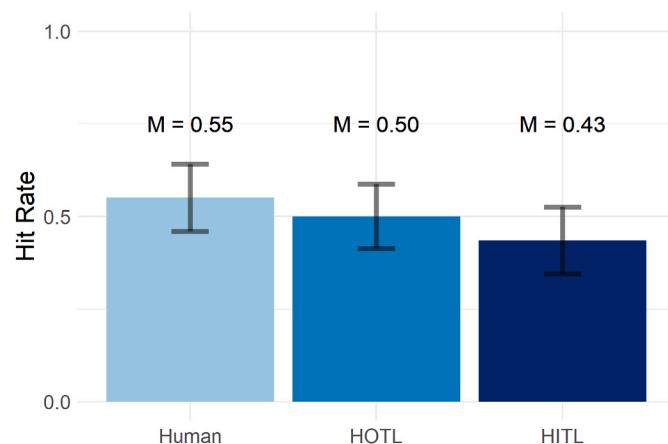
Table 3

Results of the linear mixed model for factors explaining beauty.

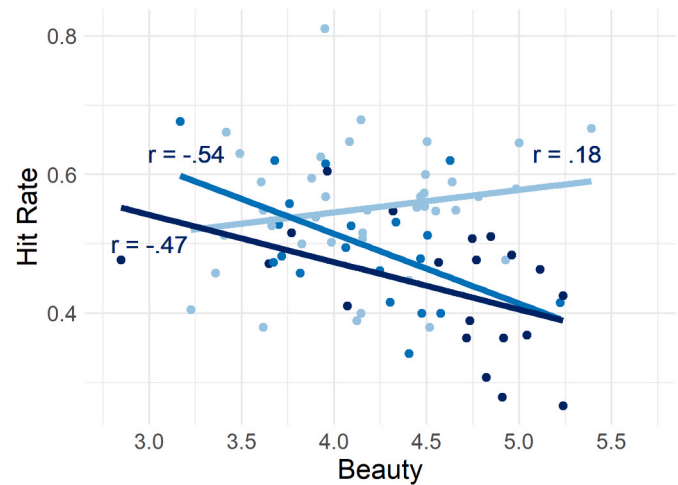
Random effects	Name	Variance	SD
ID	(Intercept)	.49	.70
Haiku ID	(Intercept)	.09	.30
Residual		.70	.84

Fixed effects	Estimate	SE	df	t value	p value
(Intercept)	4.16	.07	227.90	62.36	.00***
Déjà vu	.09	.01	14940.00	13.15	.00***
Image	.11	.01	14990.00	16.26	.00***
Valence	.05	.01	14970.00	6.32	.00***
Arousal	.04	.01	14940.00	5.05	.00***
Awe	.10	.01	14990.00	13.58	.00***
Nostalgia	.03	.01	14960.00	4.19	.00***
Novelty	−.02	.01	14940.00	−3.24	.00**
Empathy	.05	.01	14950.00	6.22	.00***
Intention	.03	.01	14950.00	3.82	.00***
Joy	.03	.01	14980.00	3.81	.00***
Continue	.16	.01	14930.00	16.83	.00***
Alive	−.01	.01	14940.00	−.56	.57
Universality	.28	.01	14990.00	32.85	.00***
Longing	−.03	.01	14930.00	−3.32	.00***
Free desire	.01	.01	14970.00	2.23	.03*
Mind wandering	.01	.01	14930.00	1.17	.24
Connection	−.01	.01	14920.00	−1.31	.19
Surprise	−.02	.01	14950.00	−2.40	.02*
Understand	.05	.01	14920.00	6.26	.00***
Tells story	.03	.01	14930.00	3.22	.00**
Task order	−.01	.11	336.50	−.08	.94
Haiku list	−.03	.10	381.00	−.25	.81
Human vs. HOTL	.11	.08	75.18	1.35	.18
Human vs. HITL	.18	.08	75.08	2.14	.04*
Education	.04	.04	381.00	1.12	.26

Note. Dummy variables were set as follows: for Task order, -0.5 for the rating first condition and 0.5 for the discriminating first condition; for Haiku list, -0.5 for list 1 and 0.5 for list 2; for Human vs. HOTL, 1 for the HOTL condition; for Human vs. HITL, the HITL condition is 1, and the other two conditions are 0.

**Fig. 2.** Hit rate in the human, HOTL, and HITL conditions.

If people believe that beautiful things are made by humans rather than AI (i.e., algorithm aversion), there may be a relationship between hit rates and beauty scores. We exploratorily examined the relationship between the beauty score and the hit rate of each haiku. The results showed positive but non-significant correlation in the human condition, $r(39) = 0.18, p = .26$, while there were significantly negative correlations in the HOTL and HITL conditions, $r(19) = -.54, p = .01$ and $r(19) = -.47, p = .04$, respectively (Fig. 3). The higher the beauty score in the AI haiku, the lower was the hit rate, reflecting that participants believed that the more beautiful a haiku, the more likely it was created by a human.

**Fig. 3.** The scatterplot between the beauty score and the hit rate.

4.4. The relationship between hit rate and personality traits

Finally, we examined the factors explaining the discrimination accuracy of authors. These factors included, the educational background, experience of haiku, and personality traits such as animism and empathy. As shown in Table 4, the higher the anthropomorphic tendency of animism ($b = .10, SE = 0.03, z = 2.81, p = .01$) and the more haiku experience ($b = 0.07, SE = 0.03, z = 2.08, p = .04$), the higher the correct response. This shows that not only the artworks' characteristics, but also the participants' factor such as anthropomorphic tendencies and experience, influenced the discrimination ability of haiku authors. Refer to Supplementary Table 3 for the exploratory results entering all personal traits measured in this study.

4.5. Exploratory analysis: the rationale behind author discrimination

We examined the relationship between the rationale behind author discrimination (what clues participants focused on to identify the haiku author) and hit rate. The most common clue for judging a haiku as human-made was "depth of the work" (72% of participants), while that for judging haiku as AI-generated was "expressiveness" (58% of participants). Multiple regression analyses with each participant's hit rate as the dependent variable showed that consistency ($b = 0.07, SE = 0.02, t = 3.91, \beta = 0.21, p = .00$) could explain the human-made haiku hit rate, while regularity ($b = 0.04, SE = 0.02, t = 2.24, \beta = 0.12, p = .03$),

Table 4

Trait factors explaining the discrimination accuracy.

Random effects	Name	Variance	SD	
ID	(Intercept)	.02	.16	
haikuID	(Intercept)	.15	.39	
Fixed effects	Estimate	SE	z value	p value
(Intercept)	.04	.05	.75	.45
IRI				
Personal Distress	.06	.03	1.88	.06
Empathic Concern	−.11	.06	−1.82	.07
Perspective Taking	.01	.03	.26	.79
Fantasy	.04	.03	1.35	.18
Animism				
Deification	.01	.02	.33	.74
Personalification	−.06	.03	−1.77	.08
Anthropomorphization	.10	.03	2.81	.01**
Aliveness Animism	.01	.01	.57	.57
Haiku Experience	.07	.03	2.08	.04*
Educational Background	.01	.02	.42	.67

N: 15400, ID: 385, haikuID: 80.

intentionality ($b = 0.09$, $SE = 0.02$, $t = 4.10$, $\beta = 0.21$, $p = .00$), and others ($b = -0.15$, $SE = 0.04$, $t = -3.39$, $\beta = -0.17$, $p = .00$) could explain the AI-generated haiku hit rate (Table 5).

5. Discussion

In this study, we evaluated haiku poetry, which has clear rules and uses a limited number of characters (i.e., conveying only limited information), created by humans and AI from various perspectives. The results showed that the AI-generated haiku which humans chose had higher beauty scores than the human-made haiku. Furthermore, the AI-generated haiku which were randomly chosen had similar beauty scores to the human-made haiku. These results suggest that human-AI collaboration will lead to better creativity and that AI's generative power is comparable to that of humans in creative fields, at least for haiku production. This study also revealed that many factors, specifically both philosophical and psychological beauty concepts, could explain the feeling of beauty in haiku. Regarding the author's discrimination, participants could not distinguish between human-made and AI-generated haiku. This result and the beauty scores discussed above, show that AI can produce work of an equivalent quality as humans. Furthermore, the personality trait of anthropomorphic tendency and haiku experience may contribute to the discrimination ability between human-made and AI-generated haiku.

5.1. The beauty score between AI-generated and human-made haiku

The results showing that human-made and AI-generated haiku without human intervention had a similar beauty score. The results are inconsistent with a previous study (Köbis & Mossink, 2021), where human-made poems were liked the most. One explanation for this might be the differences in poetic style. Haiku is the shortest form of poetry worldwide and needs to consolidate information according to a set of rules. Thus, even in this context, the quality of AI art is well ensured. It should also be noted that AI training and training materials were probably better than those used in previous studies. Haiku may be characterized by clear rules such as the 5-7-5 syllable format and the inclusion of seasonal words, which may have made training easier. Moreover, methodological differences might explain the discrepancies between our results and those of previous studies (Köbis & Mossink, 2021), in which the first two lines of the AI-generated poems were the same as the human-made ones, both poems were placed together, and participants were asked which they preferred (i.e., two-alternative forced choice). Two-alternative forced choice is relatively sensitive to subtle differences between the two poems compared with the absolute rating we employed in this study. Even if this were the case, the differences between human-made and AI-generated works are subtle. This study showed that the advantage that human-made artworks had

disappeared for non-experts when there was a slight difference in methodology.

5.2. Factors explaining beauty

We identified several factors explaining the beauty and hit rate of human-made and AI-generated haiku. One of the limitations of previous studies was that they compared human poetry with AI poetry using limited items such as likeability or beauty. However, this study overcame this limitation. For instance, the vividness of mental images, positive emotions, and higher-order emotions such as awe and nostalgia, which have been known to explain the beauty in human haiku (Hitsuwari & Nomura, 2022b; 2022c), were found to be explanatory factors for beauty in AI haiku as well. In addition, philosophical factors considered to explain beauty by philosophers and thinkers since long before the empirical study of beauty began (Brielmann et al., 2021) were again found to relate to the beauty of poetry.

5.3. Author discrimination of AI-generated and human-made haiku

The failure to discriminate between human-made and AI-generated poetry is in-line with previous findings (Hopkins & Kiela, 2017; Lau et al., 2018; Schwartz, 2015). In another study, Köbis and Mossink (2021) found that the discrimination accuracy between human-made and AI-generated poems was above the chance level, due to the high accuracy for randomly chosen AI poems (i.e., HOTL; see Table 1), suggesting that it was hard to identify AI poems with human intervention. As with other poetry genres, a major factor behind the failure to discriminate could be that the technology for the haiku generation is improving (Ito et al., 2018; Kawamura et al., 2021; Yokoyama et al., 2019). Moreover, the unique characteristics of haiku, such as being much shorter and image-dependent than typical poems, may have hindered their discrimination even in AI-generated haiku without human intervention.

Considering the negative relationship between beauty scores and hit rate in AI-generated haiku, people may have algorithm aversion (Burton et al., 2020). Algorithm aversion is an important concept in the research context of comparing human and AI poetry (Köbis & Mossink, 2021). Here, the thinking may be that humans created the beautiful haiku, i.e., they condescended AI haiku. This study is the first to suggest a relationship between algorithm aversion and author discrimination, and the result would reflect modern people's attitudes toward AI art.

5.4. Effects of experience, personality, and clues on author discrimination

Factors like the personality traits of anthropomorphic tendency and empathic concern have been examined in AI and robot interaction studies (e.g., Darling et al., 2015; Okanda et al., 2019). This study

Table 5
Rationale behind author discrimination and hit rate.

	Human						AI					
	Mean	Estimate	SE	t value	Beta	p value	Mean	Estimate	SE	t value	Beta	p value
(Intercept)		.56	.02	29.64		.00***		.46	.02	24.23		.00***
Rhythm	.30	.02	.02	1.41	.07	.16	.28	-.02	.02	-.98	-.05	.33
Consistency	.25	.07	.02	3.91	.21	.00***	.41	.00	.02	.26	.01	.79
Regularity	.06	.03	.03	.96	.05	.34	.38	.04	.02	2.24	.12	.03*
Repeatability	.02	.09	.06	1.57	.08	.12	.08	.01	.03	.26	.01	.80
Complexity	.30	-.03	.02	-1.62	-.08	.11	.13	.02	.02	.70	.03	.49
Depth	.72	-.03	.02	-1.90	-.10	.06	.23	-.02	.02	-.97	-.05	.33
Abstraction	.25	.01	.02	.53	.03	.60	.17	.01	.02	.51	.03	.61
Intentionality	.52	.00	.01	.11	.01	.91	.14	.09	.02	4.10	.21	.00***
Uniqueness	.38	.00	.01	.08	.00	.94	.05	-.05	.03	-1.58	-.08	.11
Expression	.48	-.01	.01	-.98	-.05	.33	.58	-.03	.02	-1.65	-.08	.10
Nuance	.46	-.01	.01	-.61	-.03	.54	.36	.01	.02	.68	.03	.50
Others	.03	.07	.05	1.60	.08	.11	.03	-.15	.04	-3.39	-.17	.00***

showed that they are also meaningful in the discrimination of authors in haiku works. From the results, it can be inferred that people with a high anthropomorphic tendency do not impulsively judge beautiful haiku as human-made, resulting their hit rate to be high.

Furthermore, we found that participants who paid attention to consistency—whether the words used in haiku are consistent with each other—had higher hit rates for human-made haiku than AI-generated haiku. Compared with this, participants who paid attention to regularity and intentionality had higher hit rates for AI-generated haiku. AI occasionally generated haiku that contained inconsistent meanings and were difficult to understand. Therefore, attention to consistency acted as a tool for identifying human-made haiku. In addition, regularity is often recognized in the appreciation of AI art (cf. Chamberlain et al., 2018), and the presence (or absence) of intentionality is one of the major topics in AI art research (Chamberlain et al., 2018; McCormack et al., 2019). Although the relationship between these factors and AI art detection has been pointed out, it has not been demonstrated. This study is the first to show this as data.

5.5. Limitations and future research

In this study, human-made and AI-generated haiku were examined in terms of rating and discriminating, and the relationship between beauty scores and hit rate and various variables was clarified. However, it had some limitations. One of them was the selection method of AI haiku for the HITL condition. In this study, the selection was made by three non-experts. As most participants were non-experts, it may have been easier for them to understand the HITL haiku than the human-made haiku, which experts selected. In the future, experiments with human-made haiku selected by the non-experts or HITL haiku intervened by the experts could be performed. At the same time, novices and the haiku professionals could rate the beauty or discriminate the authors as participants. Nevertheless, at least on laypersons' perspectives, this study indicates that human–AI collaboration has the potential to generate great work.

In this study, only Japanese haiku were presented to Japanese participants. AI models of haiku generation are being refined in other countries (Hrešková & Machová, 2018; Wong, Chun, Li, Chen, & Xu, 2008). Moreover, there was no significant difference in beauty scores in an Eastern-Western cultural comparison of haiku (Hitsuwari & Nomura, 2022c). Hence, this study may be applicable to other cultures as well.

5.6. Conclusions

Aesthetic evaluation and author discrimination of AI-generated and human-made haiku were conducted. The beauty scores were the same across the human-made haiku and randomly selected AI-generated haiku. However, even AI-generated haiku had the highest beauty scores if they were chosen by humans. Furthermore, regarding discrimination between human-made and AI-generated haiku, it was difficult for the participants to discriminate between them. In addition, we found a negative correlation between beauty scores and discrimination performance in AI haiku. Overall, these results suggest that in haiku (where information is minimal) the quality of AI art has reached a level which is comparable to that of humans, and the collaboration between humans and AI can produce more creative artwork (Booten & Gero, 2021). The finding suggests that creativity can be promoted by having AI assistance, which may impact art and other domains related to creativity.

CRedit author statement

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Declaration of competing interest

We have no known conflict of interest to disclose.

Data availability

All data, materials, and scripts are available online (<https://osf.io/s6wny/>).

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.chb.2022.107502>.

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